BIRD EXPERIMENT

Author: ARYA BASAK (23MS233)

Partner: Soham Bhattacharya (23MS233)

ACKNOWLEDGEMENTS

EXPERIMENT B1

Performing Experiment: Arya Basak (23MS233)

Data Collection: Soham Bhattacharya (23MS237)

Data Analysis: Arya Basak (23MS233)

Graphs: Arya Basak (23MS233)

Report Writing: Soham Bhattacharya (23MS237)

EXPERIMENT B2

Performing Experiment: Arya Basak (23MS233)

Data Collection: Soham Bhattacharya (23MS237)

Data Analysis: Soham Bhattacharya (23MS237)

Graphs: Arya Basak (23MS233)

Report Writing: Arya Basak(23MS233)

EXPERIMENT B3

Performing Experiment: Soham Bhattacharya (23MS237)

Data Collection: Soham Bhattacharya (23MS237)

Data Analysis: Arya Basak (23MS233)

Graphs: Arya Basak (23MS233)

Report Writing: Arya Basak(23MS233)

OVERALL AIM: Studying diversity, evolution and natural selection in case of birds

INTRODUCTION

Evolution and natural selection are fundamental concepts in biology that help explain how species change over time.

EVOLUTION

Evolution is the process by which living organisms change and diversify over generations. It is a fundamental principle in biology and is the reason for the vast diversity of life on Earth. The key points to understand evolution are:

- **Descent with Modification:** Species are not static entities but are continually changing over time. New species can arise from existing ones through a process of descent with modification.
- **Common Ancestry:** All living organisms on Earth share a common ancestry, meaning they are related through a long chain of ancestral species. This idea is often represented as the "Tree of Life."
- **Genetic Variation:** Genetic variation exists within populations of a species due to mutations, genetic recombination, and other factors. This variation forms the raw material for evolution.
- **Selection Pressure:** Environmental factors, such as competition for resources, predation, and changing habitats, create selective pressures. These pressures can favor certain traits and characteristics over others.

NATURAL SELECTION

Natural selection is the primary mechanism through which evolution occurs. It was first proposed by Charles Darwin and Alfred Russel Wallace in the mid-19th century. The main concepts of natural selection include:

- **Variation:** Within a population, there is genetic variation. Individuals have different traits, which can be inherited.
- **Competition:** Resources in an environment are limited, leading to competition among individuals for survival and reproduction.
- **Differential Survival and Reproduction:** Some individuals possess traits that make them better adapted to their environment, giving them an advantage in surviving and reproducing. These advantageous traits are passed on to the next generation.
- **Adaptation:** Over time, the traits that confer a survival advantage become more common in the population, leading to adaptation to the specific environment.

In summary, evolution is the overarching process of change in living organisms over time, while natural selection is the specific mechanism within evolution that drives adaptation by favoring certain traits over others based on their suitability to the environment. Together, these concepts explain the diversity and complexity of life on Earth.

For this project, 2 features of birds were studied for understanding evolution and natural selection:

- Beak Shape:
  a) Straight
  b) Round

- Body Design:
  a) Folded Wings (FW)
  b) Short Tail (ST)
  c) Crumpled Bird (CB)
  d) Short Bird (SB)

3 experiments were conducted in order to study these features.

Paper planes and cloth clips of suitable design were used for studying the effect of evolution and natural selection on body design and beak shape of birds respectively.

EXPERIMENT B1

MEASURING FLIGHT DISTANCE

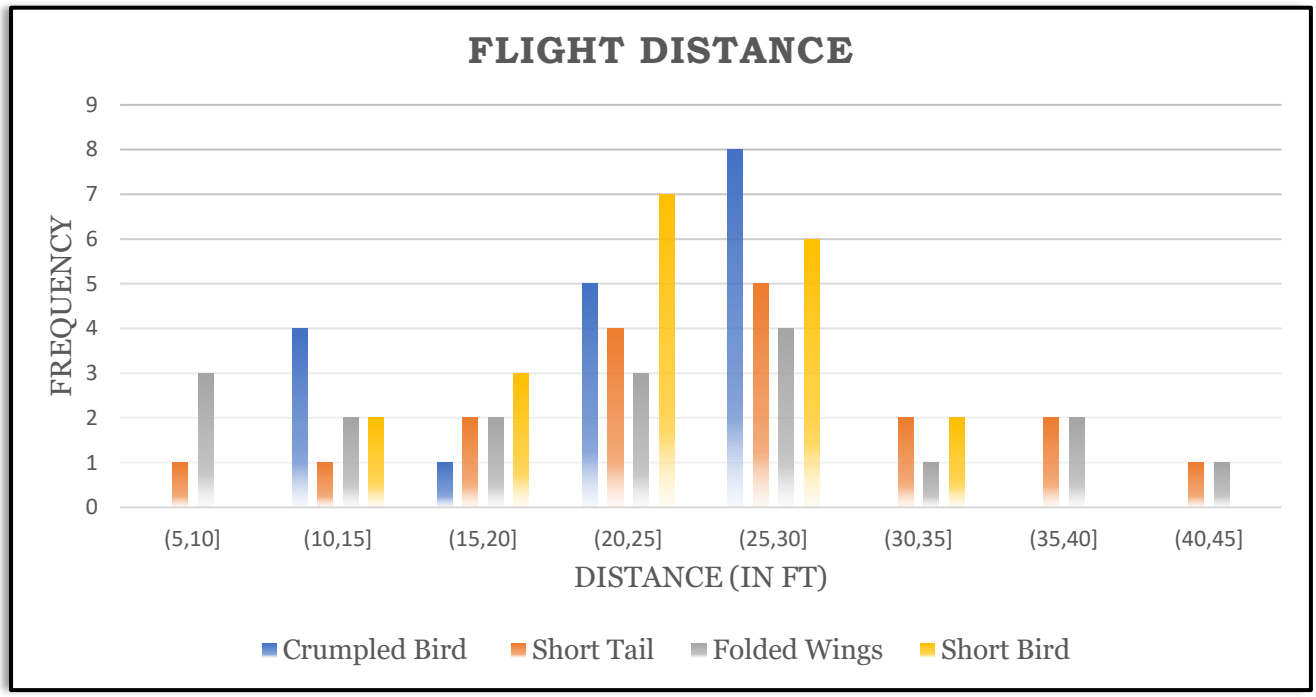
OBJECTIVE: Measuring the distance travelled by birds with different body designs

METHOD:

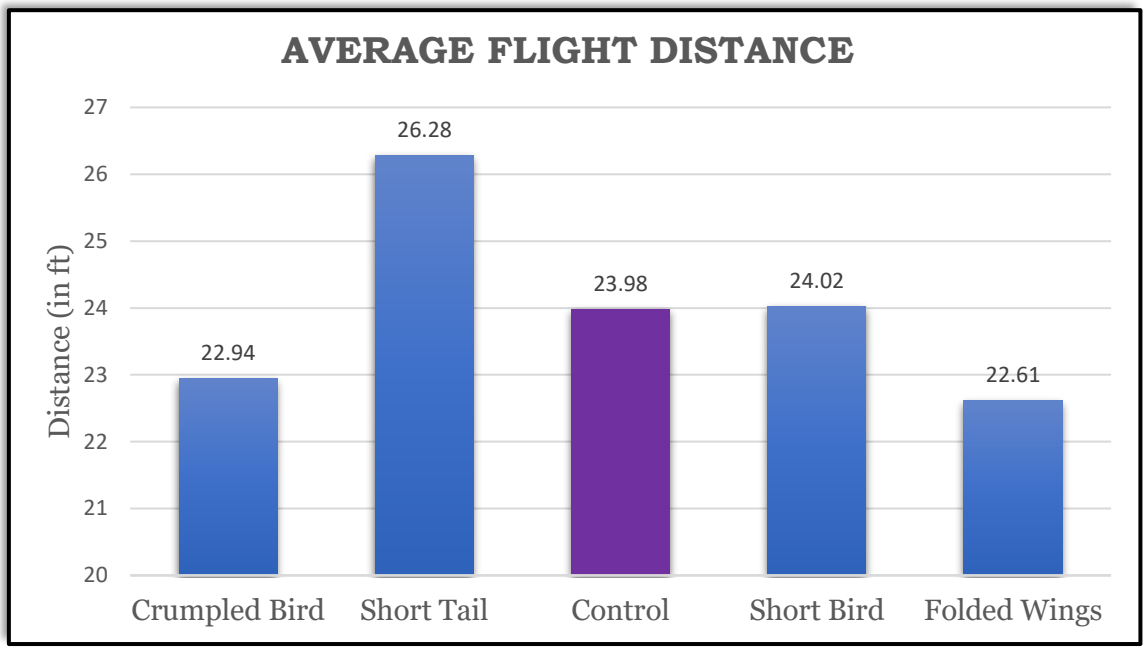
1) 6 paper planes were made for conducting the experiment – 3 of them had the same design for all the islands and groups conducting the experiment. They were termed as "Control", which represented the parent birds.
2) The next 3 planes were of different design for different groups. They represented the mutations or evolutionary changes that occur in birds. There were 4 mutations for body design – Folded Wings (FW), Short Bird (SB), Short Tail (ST) and Crumpled Bird (CB).
    a. **Folded Wings (FW):** The wings of the "Control" birds were folded once more to prepare this mutation.
    b. **Short Bird (SB):** The length of the paper used for making the "Control" birds was trimmed by 2 inches. The rest of the design was similar to that of the parent birds.
    c. **Short Tail (ST):** After making the "Control" birds, the tail of the paper planes was trimmed by 2 inches to make this mutation.
    d. **Crumpled Bird (CB):** The paper was crumpled before making this mutation. The rest of the design was similar to that of the parent birds.
3) The "Control" birds were first flown and the distance travelled by the 3 paper planes were measured separately.

4) The mutated birds were flown afterwards and the distance travelled by 3 of them were measured separately.
5) Both the "Control" and the mutated birds had to cross a certain threshold value of flight distance to be considered for the experiment.

FINDINGS:

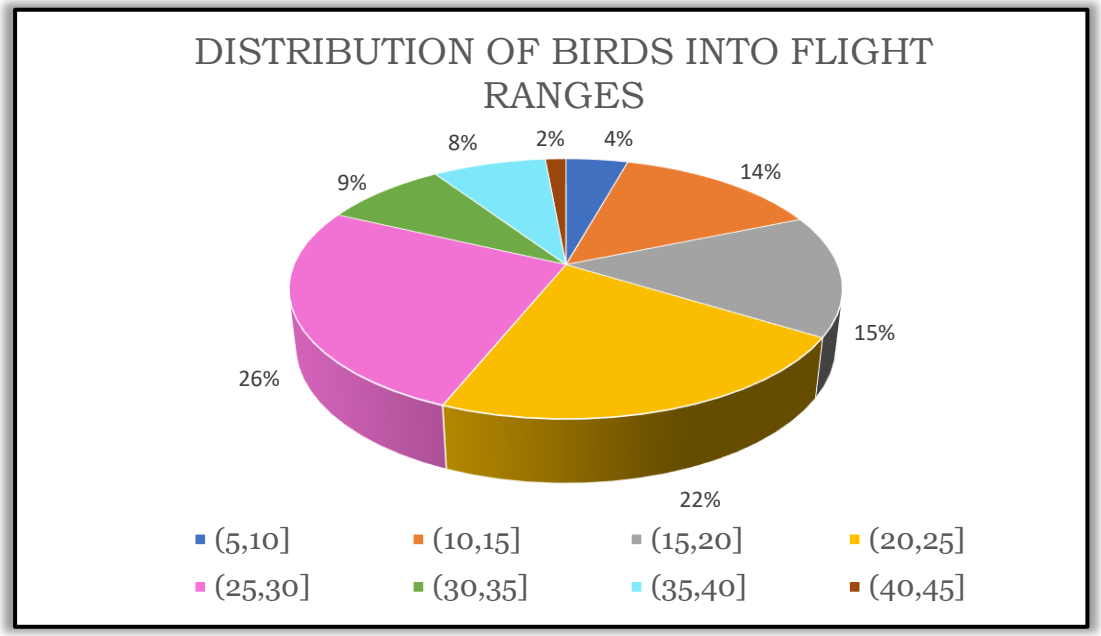


1) "Short Tail" resulted in the highest flight distance of 43 feet followed by "Folded Wings" with a distance of 42 feet.
2) As can be observed from the graph, "Short Tail" resulted in the highest average flight distance of 26.28 feet.



3) "Folded Wings" resulted in the lowest average flight distance of 22.61 feet.

4) "Crumpled Bird" and "Folded Wings" had average flight distance less than the "Control" birds, whereas others had higher than that of the "Control" birds.



5) The flight ranges of 25-30 and 20-25 feet had the highest frequency of birds (all types combined).

QUESTIONS:

1) **How will the individuals with the highest flight distance affect the population**?

   As can be seen from the chart below depicting the average flight distance for the control and mutated birds, the "Short Tail" mutation has the highest average flight distance. This is even higher than the "Control" (parent) birds.

   This means that this mutation will be selected by nature offering greater survival chances and higher reproduction rates than others. The bird population will slowly evolve with more individuals having this trait than others.

2) **Why is the flight range of 20-30 feet the most common among individuals?**

   This can be due to a variety of reasons:

   1) Environmental Factors: Topography and climate can restrict the flight range of individuals.
   2) Predator Avoidance: Individuals may fly only within their territory to avoid predators.
   3) Resource Availability: Resources may be available only within a certain range.
   4) Energy Conservation: Flying is energetically expensive; thus they conserve energy by flying only on requirement.

EXPERIMENT B2

BIRD FEEDING

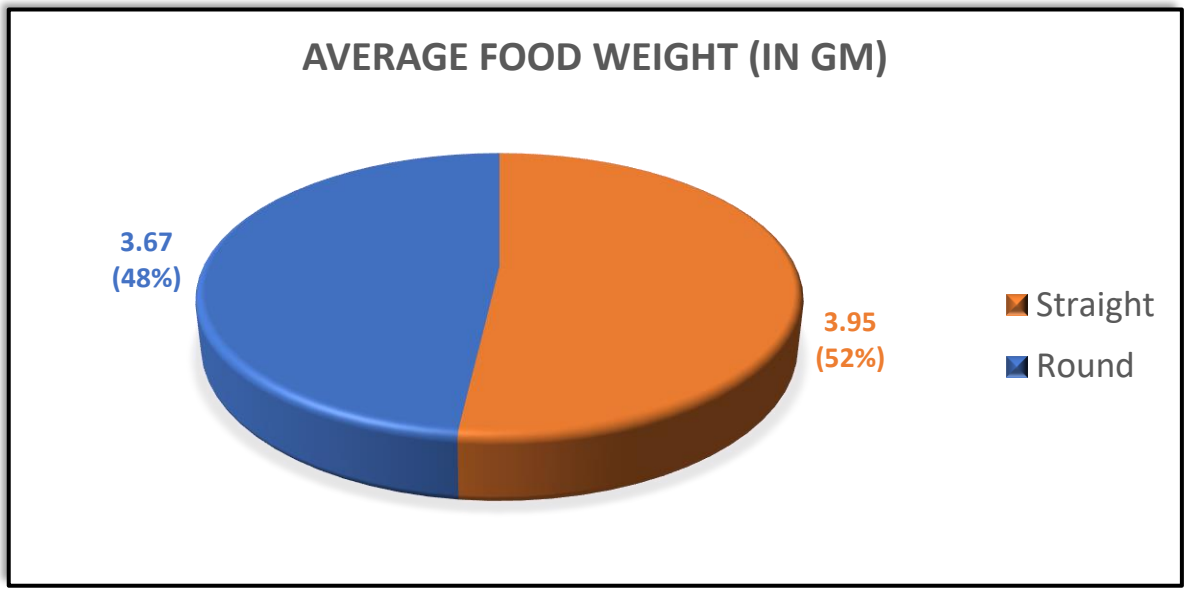
OBJECTIVE: Understanding the Theory of Natural Selection by observing the effect of mutations in beak shape on feeding ability.

METHOD:

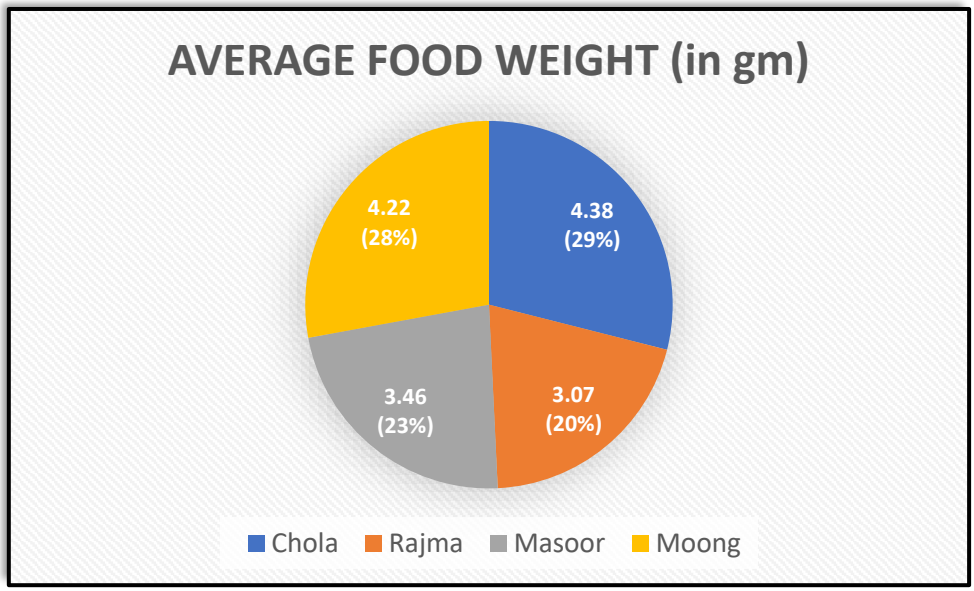
1. 2 cloth-clips; one rounded, representing a round beak and another straight, representing a straight beak of a bird were taken.
2. Different grains – chola, rajma, moong and masoor were chosen as food sources for the bird.
3. The grains were picked up with both types of clips within equal time periods allotted for both types.
4. The food grains were considered for measuring weight of food while sand particles (poison) picked up along with the food grains were deducted from the total weight.
5. Both beaks had a predetermined threshold. The bird is considered to have survived if and only if the threshold is crossed.

FINDINGS:

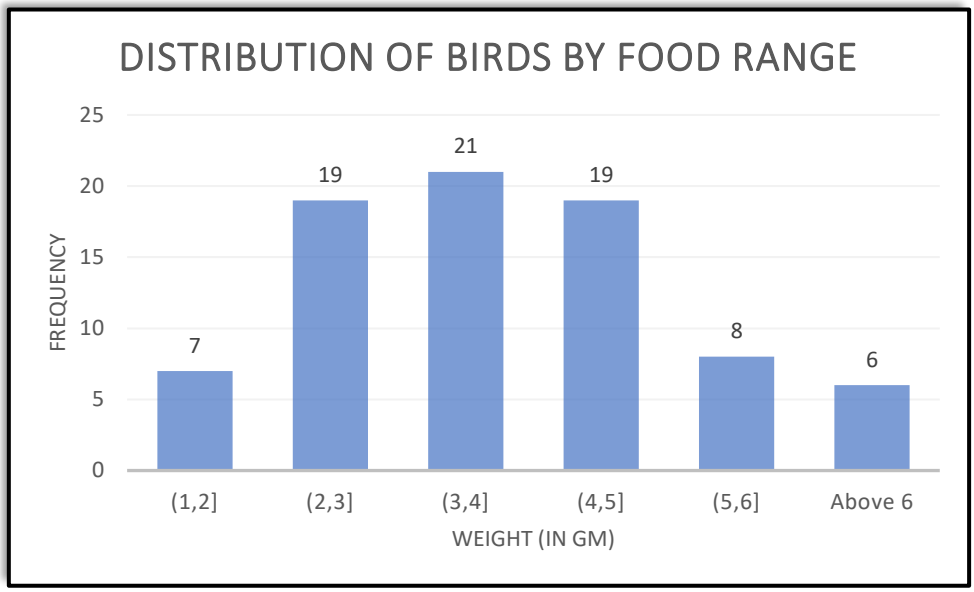
1) The "Straight" beak has a higher food weight average (3.95 gm) than "Round" beak (3.67 gm) within the stipulated time.



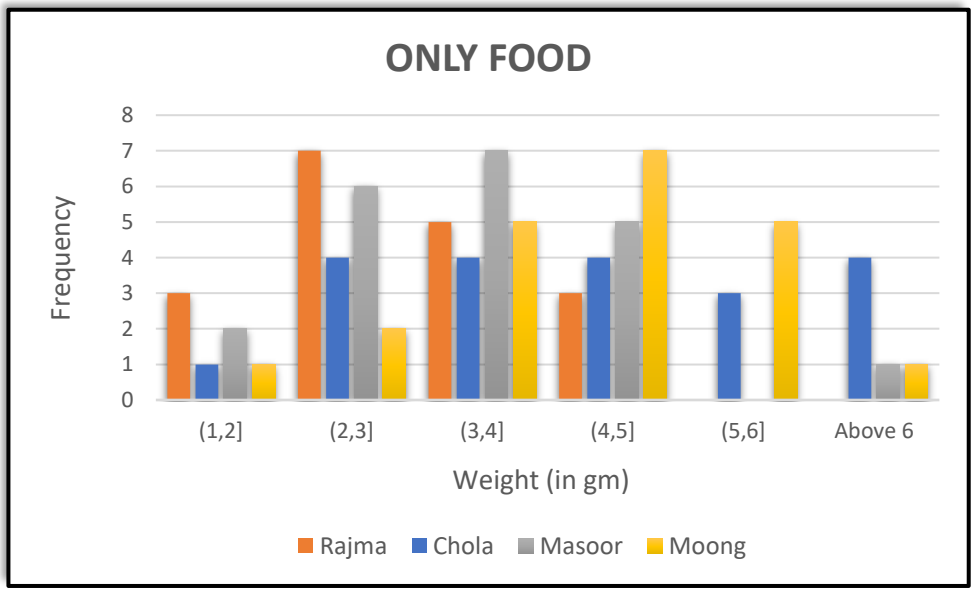
2) It is easier for birds to eat moong and chola than other food items within the specified time.



3) Most of the birds have been able to eat food in the range of 2-5 gm by weight.



4) Birds are not able to pick up rajma in larger quantities than other food items as depicted in the chart below.



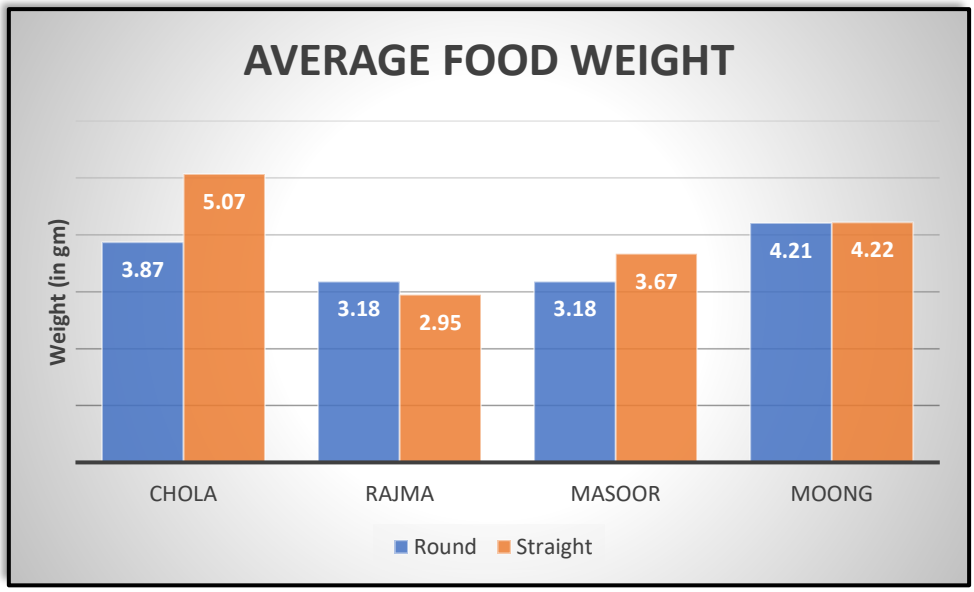
QUESTIONS:

1) **Why is straight beak able to pick up more food than round one? Does it depend on the type of food?**

   Straight beak has a higher average than the round beak. But it is only true for certain types of food.

   From the below graph, it can be seen that round beak has higher food pickup average for rajma. But straight beak dominates in all other food items. That is why it has higher average than round one.

   Overall, every bird species has evolved to possess the type of beak that is best suited for its type of food. It would be wrong to say that a beak type is more favored just because it has a higher average.



2) **Why are birds not able to pick up rajma as compared to other food items?**

   This may be due to the fact that rajma grains are bigger than other food items. This causes greater difficulty for the birds to pick it up as it may not fit inside their beaks easily.

   Other food items, being smaller, are easy to pick up as it is suitable for their beak size.

EXPERIMENT B3

COMPETTION

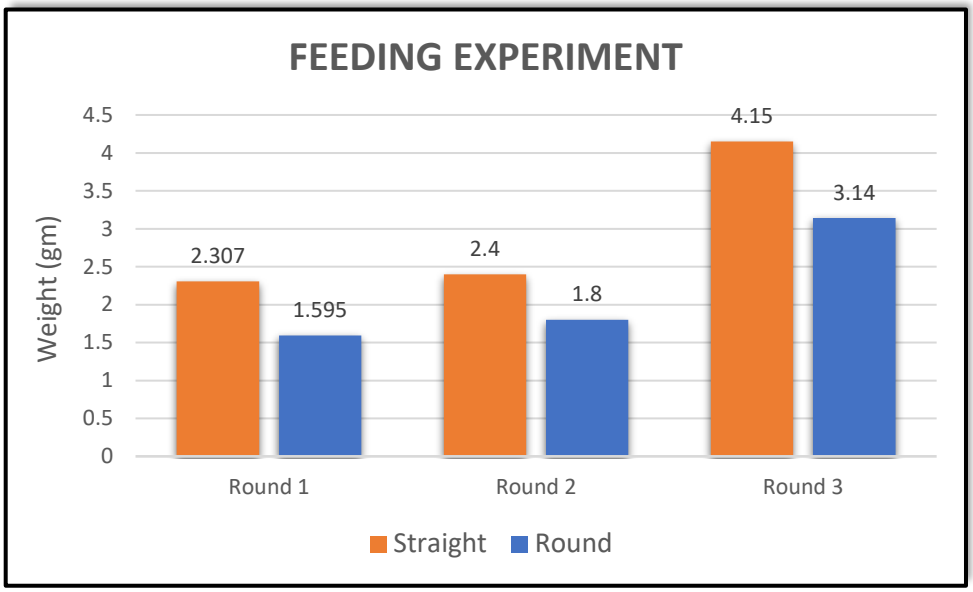
OBJECTIVE: Repeat the experiments B1 and B2 but introduce competition among the individuals

METHOD:

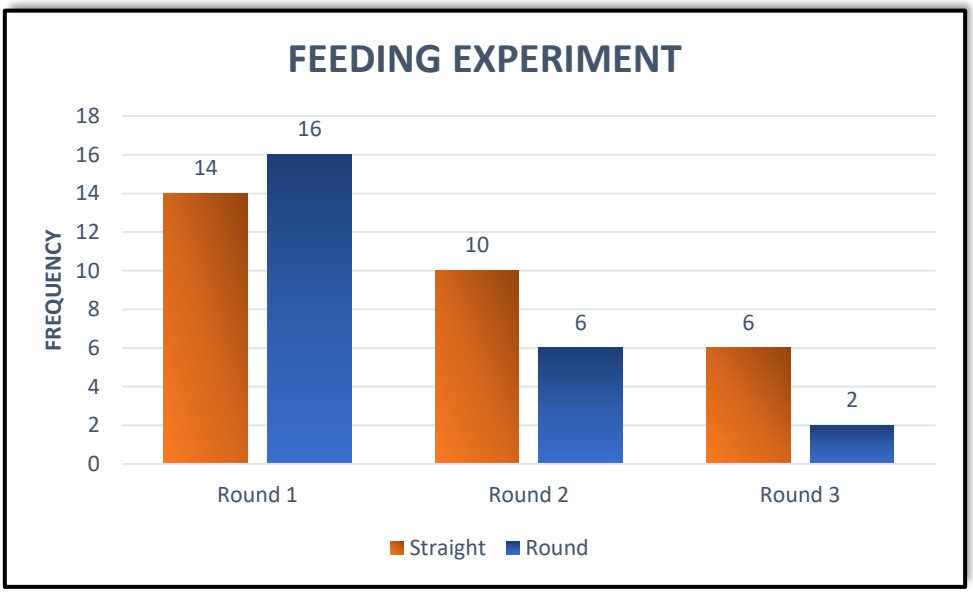
1. The bird flying experiment was conducted in much the same way as it was done in B1. Only difference is that this time, 5-6 individuals would fly at the same time. This increases the chances of collision and thus, there is competition for flight.
2. The bird feeding experiment was conducted in much the same way as was done in B2. This time though, 2-3 individuals feed on the same amount of food at the same time. There is thus competition for food.
3. In both these experiments, there is a minimum threshold that needs to be crossed in order for the individuals' performance to be recorded.
4. Both these experiments consist of elimination. From the 1st round, 4 teams get selected; from the 2nd round, 2 teams get selected and finally, 3rd round decides the winner.

FINDINGS:

1) In the feeding experiment, the "Straight" beak type dominates over "Round" beak in all the rounds, as can be seen from the chart below.

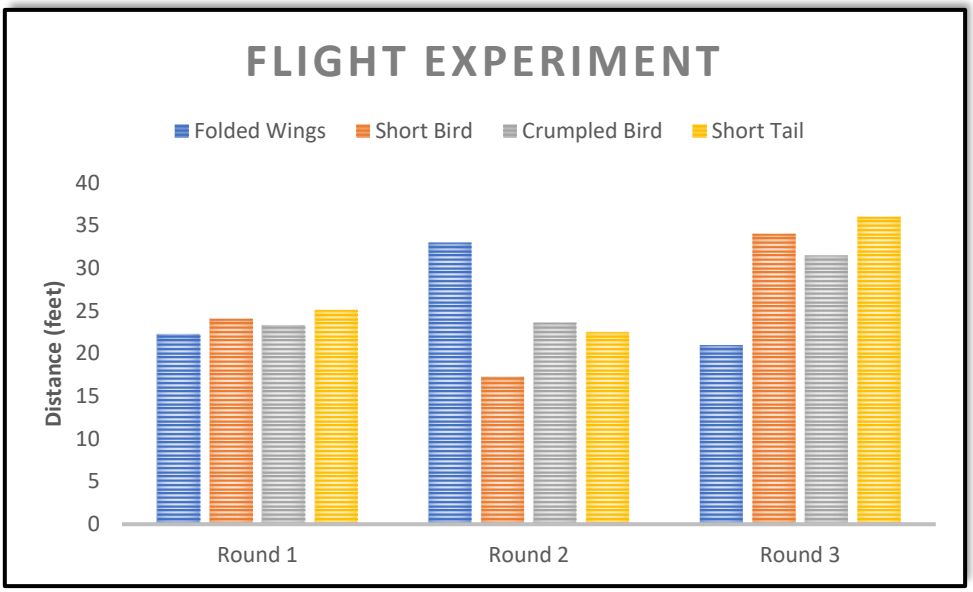


2) The number of individuals qualifying for the subsequent rounds is also higher for "Straight" beak, even though "Round" beak had higher individuals for round 1.

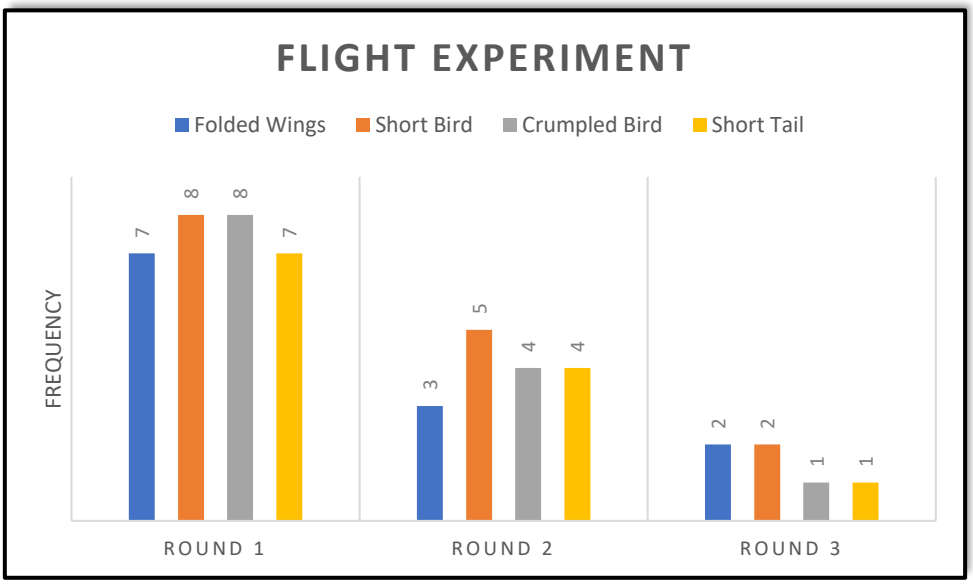


3) In the flight experiment, all mutations performed equally well in round 1; but Folded Wings had the best flight distance in round 2.

4) In round 3 however, Folded Wings flew the least distance. Short Tail had the highest average, although others performed equally well.



5) The number of individuals of each mutation qualifying for the subsequent rounds was nearly equal, as can be seen from the chart below.



QUESTIONS:

1) **Why did "Straight" beak dominate over "Round" beak? Does this indicate that it is a better mutation?**

The domination of "Straight" beak might be due to a variety of reasons:

- Competition: "Straight" beak type is more suited for a competitive environment, where there is stiff competition among individuals for limited food.
- Food Type: The type of food selected for the competition might have been more suitable for "Straight" beak type.

This does not indicate that "Straight" beak is always a better mutation. Different mutations are adapted for different conditions. This just indicates that, in the conditions in which the experiment was conducted, "Straight" beak was a better mutation.

2) **Why did Folded Wings perform the best in round 2 but the worst in round 3?**

This might be due to the fact that the Folded Wings mutation is better adapted to survive in a moderately competitive environment.

In round 1, there was stiff competition, in round 2 there was moderate and in round 3 the least, as they were elimination rounds. Folded Wings performed the best in moderate competition, but it may not be adapted to survive in less competition as other mutations will outperform it.

DISCUSSION

Close interactions with Darwin's Theory of Natural Selection over the course of this experiment has helped us develop a better understanding of this theory of utmost importance in the world of Evolutionary Biology.

'Accidents' occurring during normal chemical transactions of DNA especially during replication cause mutations. Competition between different mutations causes a 'struggle for survival' wherein the 'fitter' individuals live on to reproduce and pass on their mutations to their offsprings while the less 'fitter' ones die off along with their mutations. This causes the evolution of a species as it develops new advantageous traits suited for survival in its habitat.

CONCLUSION

From the experiments conducted we observed Straight Beak to be better adapted for feeding and Short Tail better adapted for Flight. According to Darwin's Theory of Natural Selection the offsprings will evolve to have Straight Beaks and Short Tail.

LIMITATIONS

In spite of careful considerations certain constraints aren't accounted for. Over consumption by an individual beyond its requirements may be harmful in the long run. Thus, an upper limit in the feeding experiment is required.

The same characteristic may affect several aspects of the individual's life. However, we aren't able to study them all. For a certain mutation, say small bird, its flight distance may be very little however it is able to hide itself from predators and access small confined spaces to nest or feed a lot easily than other mutations.

An interesting point to note is that evolution is very much dependent on the habitat. Rapid changes in habitat can adversely affect the population of a

species as the traits that were evolved might become disadvantageous. Some other traits which may have been less favorable might become more advantageous and better suited for survival.

FUTURE DIRECTIONS

- A brief study of the effects a mutation can have other than the ones observed directly from the experiments might just be the icing on the cake.
- A brief study of how taking into consideration the type of bird species and habitat conditions may affect the favourable mutations will also make the experiment extremely interesting.